Design for Material Jetting 302

Welcome to the Tooling University. This course is designed to be used in conjunction with the online version of this class. The online version can be found at https://www.toolingu.com. We offer high quality web-based e-learning that focuses on today's industrial manufacturing training needs. We deliver superior training content over the Internet using text, photos, video, audio, and illustrations. Our courses contain "roll-up-your-sleeves" content that offers real-world solutions on subjects such as Metal Cutting, Workholding, Materials, and CNC with much more to follow. Today's businesses face the challenge of maintaining a trained workforce. Companies must locate apprenticeship programs, cover travel and lodging expenses, and disrupt operations to cover training needs. Our web-based training offers low-cost, all-access courses and services to maximize your training initiatives.

© Copyright 2021 Tooling University, LLC. All rights reserved.

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

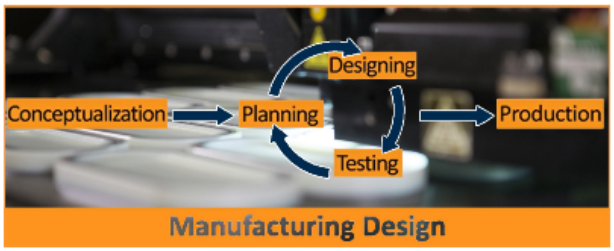
Class Outline

Material Jetting and DFAM
The Material Jetting Process
Material Jetting Materials
Material Properties for Material Jetting
Material Jetting Modulation
Material Jetting Process and Material Review
Material Jetting Design Software
Glossy and Matte Surfaces in Material Jetting
Design Issues for Material Jetting
Design Factors for Support Structures
Design Factors for Internal Structures
Finishing and Post-Processing
Final Review

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

Lesson: 1/13

Material Jetting and DFAM



A benefit of material jetting is its ability to deposit multiple types of material in a single print run to create realistic-looking objects. For example, material jetting can use multiple materials to build a single part or multiple separate parts on one build tray. The material jetting process can also blend raw materials to digitally create a digital material with specific qualities like color or hybrid properties like rigidity and toughness.

Although each AM method requires specific equipment to build parts, they all share the same basic five steps of the design for additive manufacturing (DFAM) process: conceptualization, planning, design, testing, and production. The flexibility of material jetting and other AM methods enables engineers and other designers to consider things like part functionality, complexity, and customization when designing for AM. The planning, designing, and testing steps also offer AM designers opportunities to increase production efficiency, since creating a prototype does not require additional tooling or machine configurations.

This material is based upon research supported by, or in part by, the U.S. Office of Naval Research under award number N00014-18-1-2881 led by the National Center for Defense Manufacturing and Machining (NCDMM).

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

The Material Jetting Process

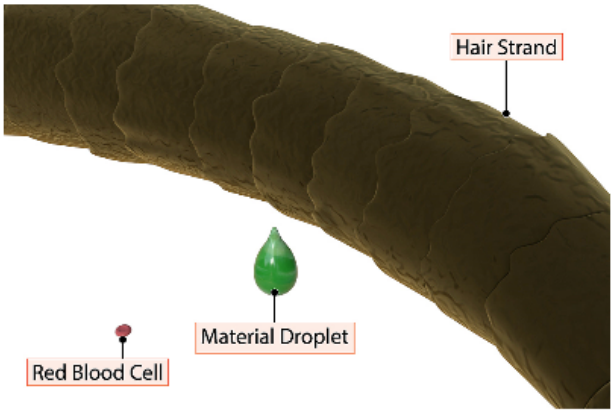
The material jetting process works much like traditional two-dimensional (2D) inkjet printers. However, instead of printing droplets of ink onto paper, material jetting dispenses droplets of build material onto a build platform to create a 3D part layer by layer. During the material jetting process, one or more print heads generally move horizontally to dispense a layer of material, while the build platform moves down vertically after the deposit of each new layer. Most material jetting processes use photopolymers, which only solidify under ultraviolet (UV) light. Because of this, most material jetting print heads have a UV lamp to flash cure the material after deposition.

Material jetting is considered one of the most accurate of the AM methods. A material jetting printhead may contain hundreds of individually controlled drop-on-demand nozzles, which dispense very tiny drops of material as small as 16 microns, or about .00063 of an inch, in thin layers to build a part. This enables the curing process to occur almost immediately when exposed to UV light. Wax, another common material used with material jetting, does not require UV curing.

Video content available online.

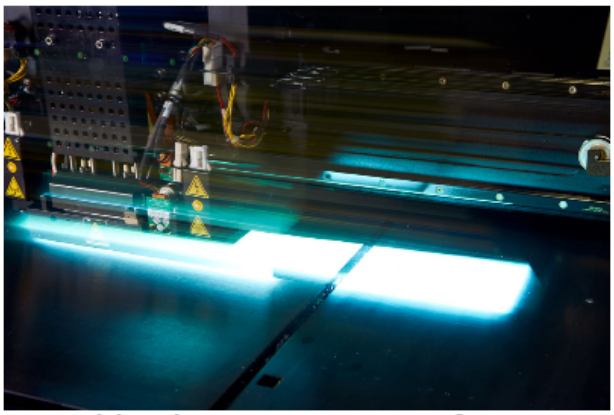
Copyright © 2021 Tooling U, LLC. All Rights Reserved.

Material Jetting Materials



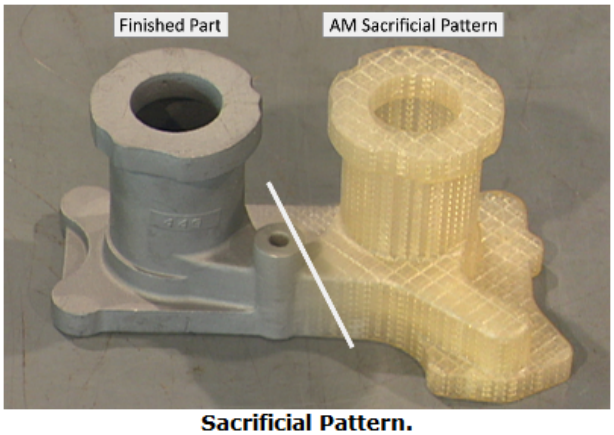
Photopolymer Droplet

Material jetting often uses a liquid-like form of thermoset resin called a photopolymer, which permanently hardens when exposed to UV light. Because of a photopolymer's sensitivity to light, the material may weaken faster over time compared to typical plastics. The fluid nature of photopolymers enables them to flow easily through the printhead nozzle as tiny droplets almost as small as a human blood cell. The droplets' small size allows the machine to deposit them very precisely on the build platform.



Material Jetting Process. Courtesy of Stratasys.

Most photopolymer types used for material jetting are proprietary, or made specifically for a certain company's machine. These photopolymers are designed to imitate popular plastics, such as acrylonitrile-butadiene-styrene (ABS) or polypropylene (PP).



Sacrificial Pattern.

The term wax refers to a variety of organic substances characterized by their high malleability and low melting point. In material jetting processes, waxes are typically made water soluble for easy removal. In material jetting, designers may use wax to create support materials and matte surfaces. Material jetting can also use wax for printing disposable or sacrificial patterns for investment casting.

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

Dental Rapid Prototype.

The ability to create inexpensive wax patterns and molds with material jetting can enable manufacturers to rapid prototype or create many prototypes in a short time. As a result, manufacturers can test and evaluate many designs and identify any possible design issues before going into a full production run. This may reduce the risk of a final part not working as expected.

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

Material Properties for Material Jetting

Hard Hat Prototype

Designers use resins in the rigid family to create stiff, precise objects such as precision tools, assembly fixtures, and prototypes that need detailed features. Designers can also use rigid resins to replicate the look and functionality of traditionally manufactured parts. For example, a material jetting machine can create a functional prototype of a product like a hard hat and other personal protective equipment (PPE). Functional prototypes enable a designer to conduct performance tests before final production.

Electrical Box Prototype

Rubber-like materials can enhance the designs of rigid parts. For example, a designer can add a rubber-like hand grip to an electric drill to make it nonslip. Manufacturers can also create near-perfect prototypes of rubber parts for testing and iteration, which is the process of creating multiple versions of a prototype to improve design. For example, a manufacturer may want to test multiple versions of an electrical junction box to determine how it reacts to extreme weather changes.

Goggle Prototype

Transparent and translucent materials can be combined with rigid and rubber-like materials to create products that have clear parts, such as

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

safety glasses and other eyewear. Designers can also use transparent materials to imitate the appearance of glass bottles or create other clear packaging.

Forceps Prototype

Many projects may require designers to customize their materials for a particular function. For example, medical devices and instruments, such as forceps, require material that is biocompatible. Other parts built with material jetting may need high temperature resistance. For instance, heat-resistant materials are necessary for testing hot air or water flow in pipes.

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

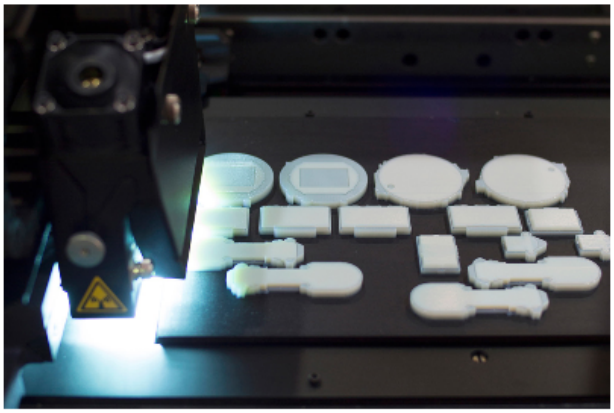
Material Jetting Modulation

Mixed Part. Courtesy of Stratasys.

Gradation enables a designer to use multiple materials at different places in the same part. This allows designers to adjust material properties or colors in specific areas of the part. A chief benefit of material jetting is that it can combine multiple materials to create a single part within the same print job, referred to as a mixed part. For example, an engineer can design the casing of a hand drill with a tough, durable plastic to absorb vibration, and combine it in the same process with a slip-resistant rubber-like material for the hand grip area.

Digital Material. Courtesy of Stratasys.

The ability to control multiple nozzles at once also enables designers to blend different build materials in their raw form, or resin state. The result of this blending creates a composite, or digital material, which gives a designer control over a parts rigidity, softness, and color. For instance, from only three primary colors, material jetting designers can create thousands of different shades of color. This enables designers to create a functional prototype with the intended color and surface finish without extensive post-processing.



Mixed Tray.

Designers can also create multiple parts with different materials in a single tray, referred to as a mixed tray. This enables a designer to create

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

separate objects, like a blue rigid plastic container and soft rubbery sole of a sneaker, during the same build. Mixed tray printing can also enable multiple designers to print their parts at the same time, leading to quicker turnaround times for prototypes and other parts.

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

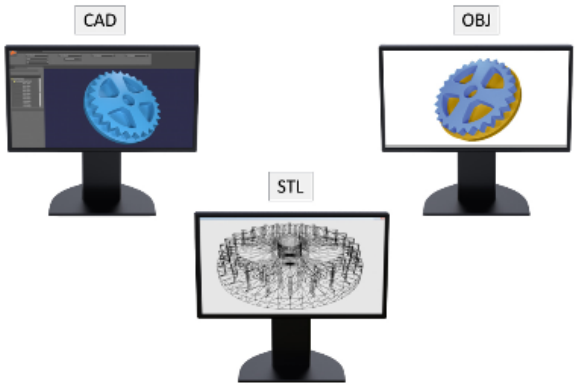
Lesson: 6/13

Material Jetting Process and Material Review

1. Material jetting machines use an extrusion head to deposit heated material on a build platform in layers. **FALSE**
2. Photopolymers require ultraviolet light to harden and solidify. **TRUE**
3. Print heads on material jetting machines have a single nozzle that can deposit different types of materials during the same print job. **FALSE**
4. Mixed tray refers to the process of creating multiple parts with different materials in a single print run. **TRUE**
5. Support materials for material jetting are typically made from waxes. **TRUE**

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

Material Jetting Design Software

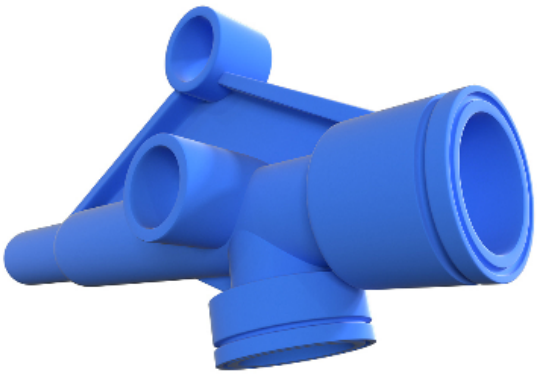


Because of this, a designer must convert the 3D design file into an STL file format. The STL software converts the geometry of the 3D model into a sequence of interconnected triangles representing the external surface of the design. If a part has a different material in a certain area of the part, then a designer must create a separate STL file for each area. If a part has blended materials, then a designer may also need to create a VRML file or an OBJ file, either of which communicates characteristics like color and surface quality to the machine.

Next, digital slicer software translates the STL, VRML, and OBJ information into digital versions of the 2D layers that the machine will deposit to physically build the part. The digital slicer software then converts the 2D layer information into G Code, a numerical programming language that controls the functions and motions of machine tools like material jetting machines. Essentially, digital slicer software breaks a digital model into the layers of material that the material jetting machine will deposit and bind, while the G code program directs the print head movement back and forth across the build platform.

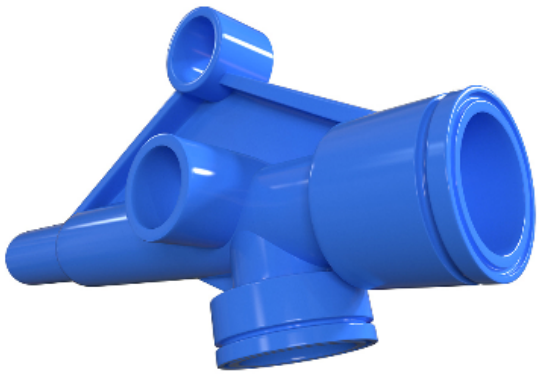
Copyright © 2021 Tooling U, LLC. All Rights Reserved.

Glossy and Matte Surfaces in Material Jetting



Matte Surface

In the matte setting, a material jetting machine will cover the entire part with a thin layer of support material, regardless of structural requirements. The process of surrounding the part with support material creates a uniform matte surface across the entire part. Typically, matte surfaces have a rough, textured appearance compared to smooth glossy finishes. Matte surfaces require more material and more post-processing, which typically makes matte surfaces more costly to produce.



Glossy Surface

Designers use the glossy setting to create a sleek, shiny object surface. Unlike the matte setting, glossy surfaces only use support material when a structure requires it, because support material may damage the part when removed. Typically, material jetting machines cannot print glossy surfaces underneath support material, which makes it extremely difficult to produce a uniform surface. To avoid this issue, designers may limit or avoid using support structures for parts with glossy finishes. Designers may also print a part in multiple pieces for later assembly, which reduces the need for structural support material with both glossy and matte settings.

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

Design Issues for Material Jetting

Material jetting has a well-earned reputation for creatin realistic-looking prototypes. However, the brittleness and low ductility of material jetted parts makes them less optimal for end-use production. In addition, photopolymers used in the material jetting process typically have a high degradation rate, which means that they break down relatively quickly when compared to other plastics. For photopolymers, using the correct UV light setting is crucial. Low UV intensity during the curing process may not fully solidify the part, which can cause sagging or a wavy surface texture. Checking the UV light before each build may help avoid this issue.

The importance of part orientation, or the position of the part as it is built, is typically reduced with material jetting. Because of the fine layering of material jetting, it reduces the likelihood that certain building positions will reveal layer lines. AM machines can build parts in a variety of different positions, all of which can affect their final quality. For example, building a part facing upward helps protect against potential defects during printing. Even in material jetting, designers must consider the orientation of any holes that may appear in a part. An incorrect support material setting on a machine may sometimes cause a matte surface to appear around holes or other openings when printing a glossy surface.

Figure 1: Part Sagging

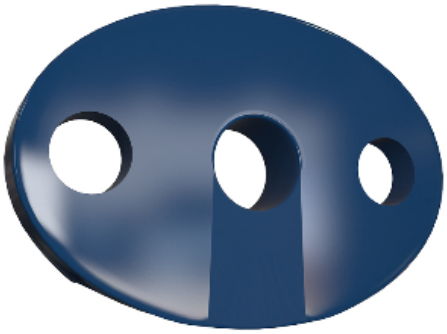


Figure 2: Matte surface appearing around holes.

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

Design Factors for Support Structures

Material jetting processes typically use the same wax material to create a matte finish as they do to form support structures, which provide reinforcement for a part during the build. Regardless of the part orientation, material jetted parts need support structures for any overhangs or under-supported sections. However, as a general design rule, limiting the need for support structures in a part can help reduce the costs associated with support material.

Material jetting support material typically comes in three categories, based on the way material jetting machine operators remove the material after printing. Operators can remove the first category of support material by hand or water pressure. A second, less popular category requires an operator to use a caustic chemical solution to dissolve the support material. This method helps provide a smoother part surface, but the required chemicals are harmful to employees and the environment. A third category of support material is completely water-soluble, which makes it optimal for parts with fine features. However, water-soluble support structures can only work with certain machines and base materials, which typically makes it the least used of the three options.

Figure 1: Courtesy of Stratasys.

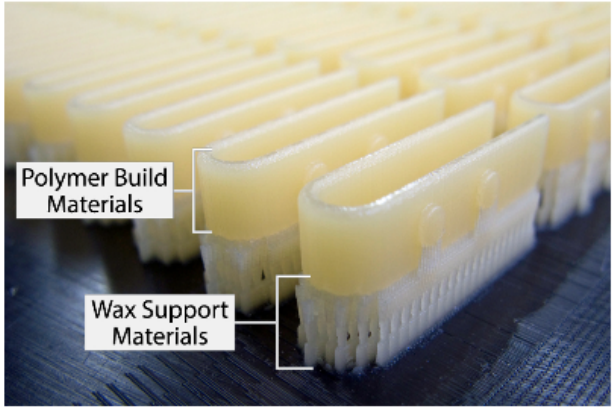


Figure 2: Material Jetting Part and Support Material

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

Design Factors for Internal Structures

Internal structures can present unique challenges when building with material jetting. For example, holes or other hollow part features may sag or distort during a part build. Part designs with hollow features therefore typically require plans to fill the empty space with support material during the build process. Removing this material from the completed part may be difficult, especially with small or extended hollow spaces. Because of this, a designer may add an escape hole to the part to make support material removal faster and easier. If an escape hole isn't feasible, a designer might extend the support material outside of the part for easier hand removal.

Designers must also consider the thickness of any walls that surround a hollow feature. Thinner part walls may deform or break under stress, especially in joint areas where material sections meet. To help provide more strength and stability, designers can add a fillet, or rounded corner, to reinforce part joints.

When designing for material jetting, engineers can adjust for changing variables of their build process, or even the physical part design itself. For example, if the part has overhanging components a print head may deposit wax as a support structure. Adding a rib or boss to the part design can help strengthen the wall of a part. However, material jetting build materials are generally more expensive compared to other AM technologies, so adding material may impact production rates by increasing material costs.

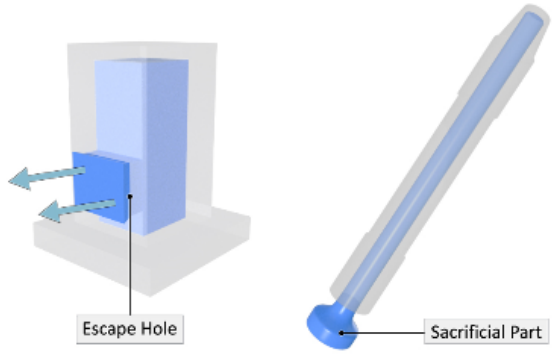


Figure 1: Design Solutions for Hollow Part Features

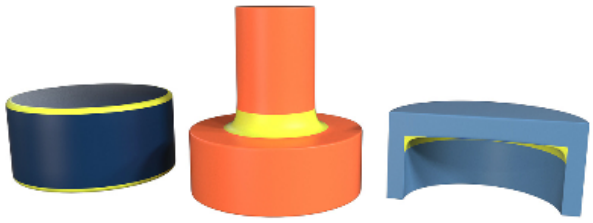
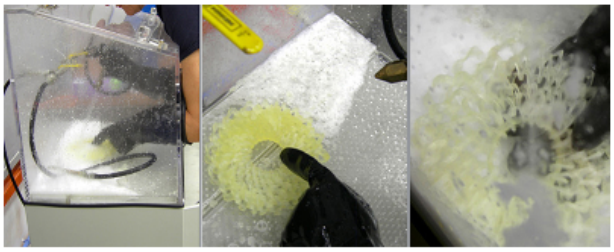


Figure 2: Fillet Examples

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

Finishing and Post-Processing



After printing, a material jetted part will typically need additional post-processing or finishing, which is a group of procedures used to clean, improve, or create an attractive surface finish on a part. Support material must generally be removed, which requires specific techniques to avoid damaging any delicate part features. For example, removing support material using water pressure requires a special chamber with operators wearing additional PPE. Before removing support structures with a caustic chemical solution, an operator will need to prepare a cleaning tank with the proper chemical ratios. Soaking a part in caustic chemicals might take hours. Because of this, operators may try to remove as much material as possible by hand before soaking to speed up the chemical removal process. For larger part runs, an operator may soak the parts in batches.

Post-processing can improve surface finish and remove blemishes or imperfections that may have occurred during the material jetting process. Because post-processing may add or remove material to a part, designers must factor it into their part design. For example, if a part needs sanding, a material jetting designer should build the part a little larger to account for the removed material. In other instances, some designers might apply a coat of clear paint to a transparent surface to give it a more vibrant finish. Instead of painting, operators may add a glycerol coating to strengthen parts and better protect them against environmental factors.

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

Lesson: 13/13

Final Review

- **STL Software**: Converts the geometry of a 3D model into a sequence of interconnected triangles.
- **Matte Surface**: Uses support material to create a rough, uneven texture across the entire part.
- **Part Orientation**: How a designer positions a part during the material jetting build process.
- **Escape Hole**: Adding this to a part can make support material removal faster.
- **Water-Soluble**: Optimal support material for parts with fine features.
- **Wax Material**: May damage the delicate features of a part when removed.

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

Class Vocabulary

3D	Three-dimensional. Having a length, depth, and width. Most physical objects are three-dimensional.
3D Printing	The process of producing a 3D object using a specialized tool that creates successive layers of material. 3D printing must technically involve the use of a print head or nozzle, but the term is often used interchangeably with additive manufacturing (AM).
Acrylonitrile-Butadiene-Styrene	ABS. A thermoplastic material with good heat, chemical, and impact resistance. Acrylonitrile-butadiene-styrene is used to manufacture medical equipment housing.
Additive Manufacturing	AM. The process of successively layering materials to make an object based on a three-dimensional computer model. Additive manufacturing allows for rapid prototyping, mass customization, and increased part complexity.
AM	Additive manufacturing. The process of successively layering materials to make an object based on a three-dimensional computer model. AM allows for rapid prototyping, mass customization, and increased part complexity.
Base	Unblended or raw. The base is essentially the foundation.
Blend	The process of mixing materials. Blending is used in material jetting to combine two base resins together.
Caustic	Capable of destroying organic or inorganic material by chemical action. Caustic substances can cause injury, which can be avoided by wearing appropriate PPE.
Computer-Assisted Design	CAD. Computer software used to create a 3D model of a part before it goes into prototyping or production. Computer-aided design models are digital representations of the different aspects of a part.
Conceptualization	In the five-step design process, this segment forms an idea about a possible part or manufactured good. Conceptualization is one of the initial stages of design for manufacturing (DFM) or design for additive manufacturing (DFAM).
Degradation Rate	The amount of time it takes for material to deteriorate. Degradation rates measure deterioration due to friction, heat, and chemical reactions.
Design	In the five-step design process, the creation of the actual part specifications. Designing involves creating the blueprints and prototypes for a part.
Design For Additive Manufacturing	DFAM. The methodology of planning, testing, and creating an additively manufactured part that functions optimally. Design for additive manufacturing allows engineers to mostly focus on part functionality.
Digital Material	A specially blended composite material created from different types of basic resins. Digital materials enable material jetting designers to control part qualities such as strength, rigidity, and color.
Ductility	The measure of a material's ability to be drawn, stretched, or formed without fracturing. The ductility of plastic depends on its molecular structure.
End-Use	Designed to be used directly by a consumer or directly in another manufactured product. End-use products created by additive manufacturing (AM) include medical implants, custom dental devices, and camera equipment.
Escape Hole	A design solution for removing excess materials from a hollow part. Escape holes are used in material jetting to safely reduce the risk of damage to a part during post-processing.
Fillet	A rounded internal corner. A fillet adds strength to the corner of a part.
Fixtures	A customized workholding device used on machine tools to position and hold a part during various machining operations. A fixture is built to hold a specific part design.
Flash Cure	To cause a material to bond and solidify by exposing it to a flash of ultraviolet (UV) light. Material jetting systems use an ultraviolet lamp to flash cure build materials as they are deposited on to a

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

jetting systems use an ultraviolet lamp to flash cure build materials as they are deposited on to a build platform.

Forceps A tweezer-like instrument used for pinching. Forceps are used in the medical field for surgery and must be biocompatible.

Functional Prototype A part model produced through additive manufacturing that represents the composition and design of the final part. Functional prototypes are used to test how a part will perform in its intended application.

G Code A programming language that uses letters and numbers to form commands for machine tools. G code programs are used to direct tool movements in additive manufacturing, CNC machining, and hybrid machining.

Glossy A sleek surface with shine. Glossy surfaces are difficult to print under support structures, especially with material jetting.

Glycerol A colorless, odorless liquid used to make polyester fibers and antifreeze. Glycerol is a protective material used to post-process material jetted parts.

Gradation Additive manufacturing process containing a series of changes in material composition. Gradation allows a single, solid part to have different materials, properties, or colors in different places.

Hand Drill A hand-held tool used to create holes or drive screws. Manufacturing hand drills require a number of different materials, including multiple types of plastic.

Investment Casting A casting process that creates intricate shapes using an expendable wax pattern and mold. Investment casting is also called precision casting and lost-wax casting.

Iteration Repeating actions and steps in processes. Iteration is an important aspect of design for additive manufacturing (DFAM) because it allows for constant and quick re-designs during initial stages.

Joint A point at which two objects or materials are connected, usually facilitating movement between the objects or materials. Joints can be directly manufactured using additive manufacturing (AM) with no additional assembly required.

Malleability Able to be formed or shaped through impacts or pressure. Malleable materials do not break or fracture easily.

Material Jetting An additive manufacturing (AM) method in which droplets of build material are selectively deposited onto a build platform. Material jetting, or PolyJetting, systems use a photopolymer that is cured by ultraviolet (UV) light.

Matte A dull surface without shine. Matte surfaces made with material jetting require an object to be covered with wax support material.

Matte A dull surface without shine. Matte surfaces require more material and more post-processing, especially with material jetting.

Melting Point The temperature at which resin becomes liquid. When a hardened resin melts to a liquid, the molecules can move about freely.

Microns A metric measurement equal to 0.000001 of a meter, or approximately 0.000039 of an inch. Micron is a shorter term for micrometer.

Mixed Part An additive manufacturing (AM) process where a combination of build materials are fabricated together to build a single part. Mixed parts are used to combine different materials for a part in one build tray.

Mixed Tray An additive manufacture (AM) process in which a single build tray contains multiple items made with different types of materials. Mixed trays are used to build a wide range of parts in a single print job.

Modulate To adjust settings like frequency or strength. Modulating is used with material jetting to adjust the types of materials used to build a part.

Nozzles A spout at the end of piping or tubing through which substances are funneled. Nozzles are used to distribute build material during some additive manufacturing processes.

OBJ Software that can create or convert files which represent the color, texture, and complex surface geometry. OBJ files are used in material jetting, similar to VRML files, to designate color or texture on a per face, or section, basis.

Part Orientation The position of an additively manufactured part during its build process. Selecting the ideal part

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

Part Orientation The position of an additively manufactured part during its build process. Selecting the ideal part orientation involves balancing part quality and build times.

Personal Protective Equipment PPE. Any clothing or device worn to minimize exposure to hazards or prevent injury. Common additive manufacturing personal protective equipment includes gloves, safety goggles, and nonflammable clothing.

Photopolymers A thermoset polymer that cures and hardens when exposed to light. Photopolymers are used in material jetting and vat photopolymerization systems.

Planning In the five-step design process, this segment assesses the best ways to create the conceptualized part. Planning involves initial considerations of the design and of the practical concerns related to the manufacturing process.

Plastic A manufacturing material consisting of large molecules, characterized by being lightweight, corrosion resistant, and having a low melting point. Plastic is usually easy to shape and form.

PolyJetting An additive manufacturing process in which droplets of build material are selectively deposited onto a build platform. PolyJetting, or material jetting, systems use a photopolymer that is cured by ultraviolet (UV) light.

Polypropylene PP. A polymer that is tough, flexible, and resistant to fatigue. Some proprietary thermosets are designed to mimic the properties of polypropylenes.

PPE Personal protective equipment. Articles of clothing or safeguarding devices that workers use to prevent injury in the workplace. Safety glasses, insulated gloves, face masks, and earplugs are examples of PPE.

Print Heads The moving component on a printer that holds and distributes the part build material. Additive manufacturing machines, such as those used in material jetting or binder jetting, have print heads.

Production In the five-step design process, the manufacturing of a finished part for delivery to a customer. Production is the final stage of design for manufacturing (DFM).

Proprietary Designed by a specific company for use only with its own systems. Proprietary materials include many polymers and superalloys that have been designed for specific applications.

Prototype A preliminary model of a part used to evaluate the look and performance of a design. Prototypes are used to determine the specifications for the final part.

Rapid Prototype A product development technique where additive manufacturing processes are used to create prototypes for a traditional manufacturing operation. Rapid prototyping allows engineers to quickly create a number of prototypes in a short time period, reducing lead time.

Raw Materials An unprocessed or lightly processed component used to make the finished part. Raw materials in manufacturing include metal, polymer, and ceramic.

Resins A raw polymer, usually in the form of beads or pellets, that is not yet molded into its final shape. Resin is melted to form plastic parts.

Rigid Resistant to bending. Rigid materials are stiff and inflexible and are useful in applications where a part must hold its shape.

Sacrificial Patterns A disposable replica of the part or casting that is used to create a negative impression in a mold. A sacrificial pattern is used in investment casting.

Slicer A computer program that divides an STL file into layers for additive manufacturing (AM). Slicers, or slicing programs, can either be separate computer programs or part of the machine interface.

Software Coded instructions or programs that control computer hardware functions and operations. Software is used to design additively manufactured parts and command machine tool movements during part creation.

Soluble A material that dissolves when exposed to a solvent, such as water or a liquid chemical. Soluble materials are often used for support during some additive manufacturing methods.

STL Software that can create or convert digital design files which represent 3D models into a series of interconnected triangles. STL software is used in additive manufacturing processes.

Support Materials Additional substances used in the additive manufacturing (AM) process to support the product as it is being constructed. Support materials are removed from the AM product as part of the additive manufacturing process.

Copyright © 2021 Tooling U, LLC. All Rights Reserved.

Testing In the five-step design process, examining a part to ensure that it performs its intended function and that it can be satisfactorily manufactured. Testing indicates whether the part needs additional planning or if it is ready for production.

Thermoset A group of polymers that are permanently hardened by heating. Thermosets typically require ultraviolet (UV) light in order to harden into their permanent shape.

Translucent A type of material that transmits light. Translucent materials typically diffuse light in the process of transmission.

Transparent Having the quality of allowing light to pass through. Transparent objects are clear and can be seen through.

Two-Dimensional 2D. Occurring only in the axes of length and width. Two-dimensional objects are flat and lack height.

Ultraviolet UV. A potentially harmful wavelength of light that is below, or shorter than, violet on the light spectrum. Ultraviolet light is used to selectively harden photopolymers in some additive manufacturing processes.

UV Ultraviolet. A potentially harmful wavelength of light that is below, or shorter, than violet on the light spectrum. UV light is used to selectively solidify build materials in vat photopolymerization and material jetting processes.

VRML Software that can create or convert files which represent the color and texture of 3D models. VRML files are used in material jetting, similar to OBJ files, to designate color or texture on a per face, or section, basis.

Water-Soluble The ability of a substance to dissolve in water. Water-soluble support material is easy to clean from parts made with additive manufacturing (AM).

Wax A variety of organic substances characterized by the ease with which they can be shaped. Waxes are sometimes used as a support material during material jetting.

Copyright © 2021 Tooling U, LLC. All Rights Reserved.